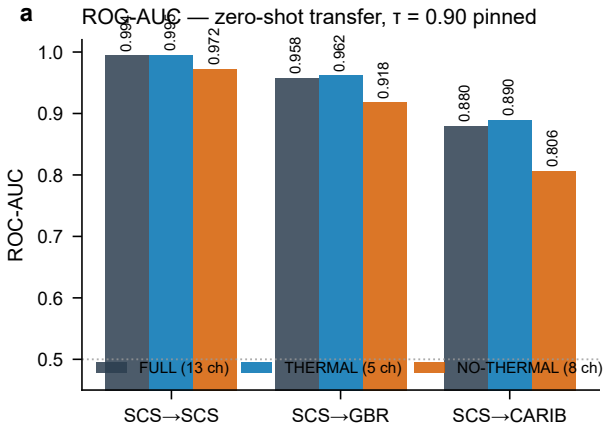
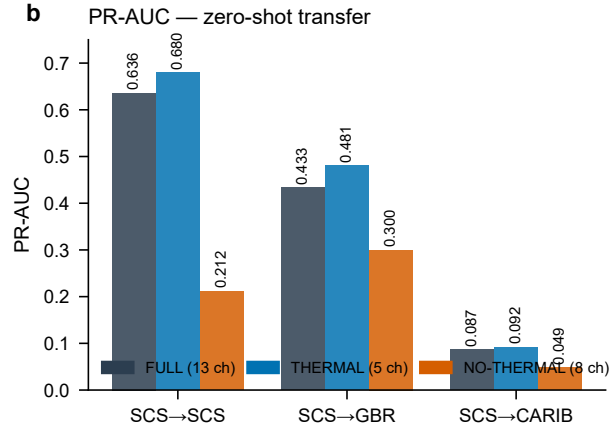
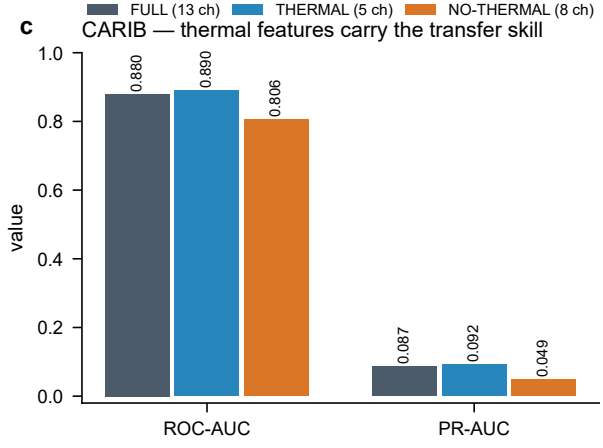
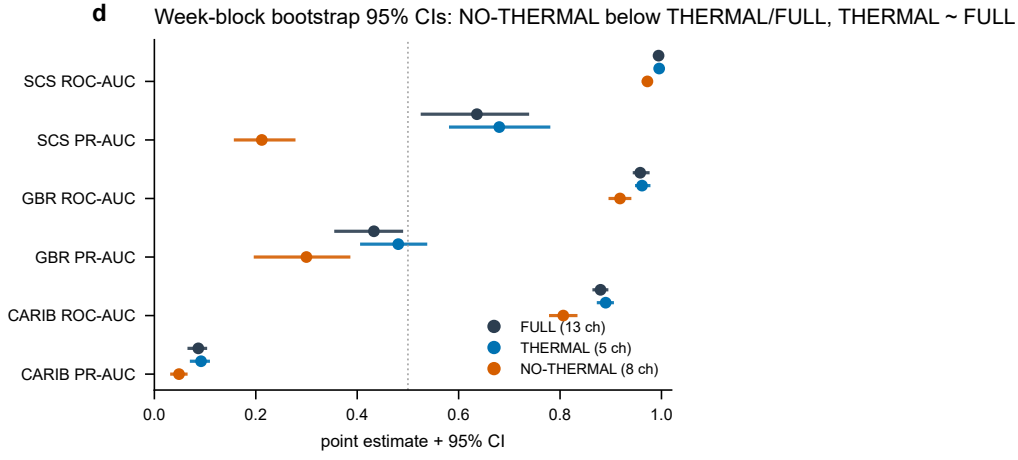
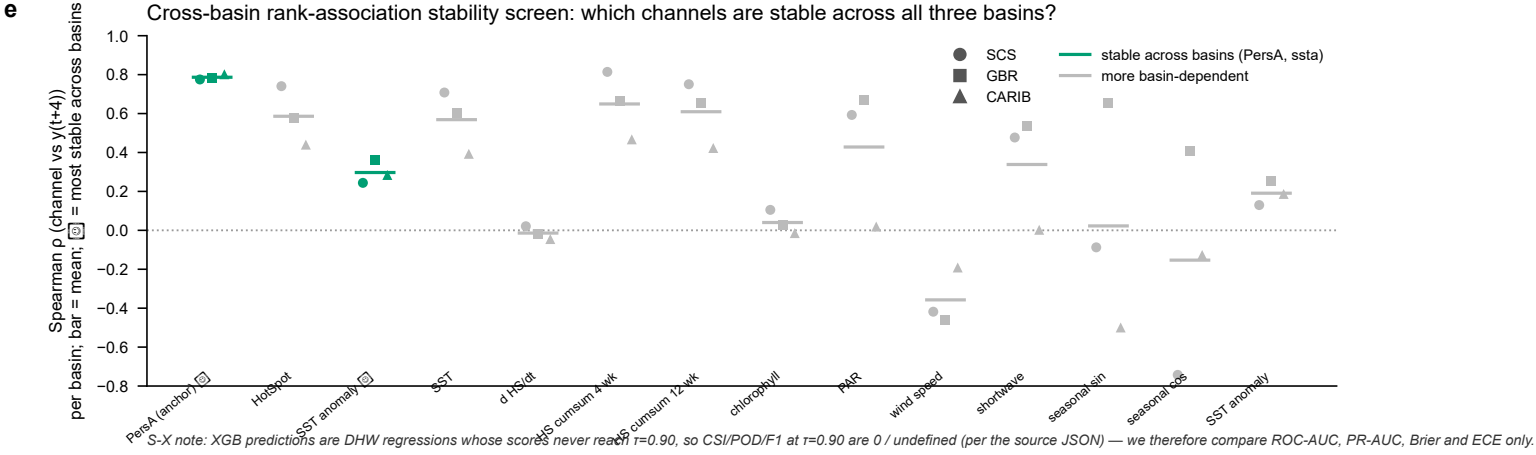










S-X note: XGB predictions are DHW regressions whose scores never reach τ=0.90, so CSI/POD/F1 at τ=0.90 are 0 / undefined (per the source JSON) — we therefore compare ROC-AUC, PR-AUC, Brier and ECE only.

XGBoost and TSCF use the same SCS→GBR/CARIB zero-shot transfer protocol: lead=4 wk, history=24 wk, SCS-train-only scaler, τ=0.90 pinned; GBR/CARIB never enter training, tuning or threshold selection.